Method	Layers	Nodes	Dropout	α	σ
Cox-MLP (DeepSurv)	6	256	0.1	-	-
Cox-MLP (CC)	6	128	0	-	-
Cox-Time	8	256	0	-	-
DeepHit	6	512	0.1	0.001	0.5

Table 6: KKBox model architectures. α and σ only applies the DeepHit (see Appendix D).

Method	C^{td}	IBS	IBLL
Cox-SGD (Linear)	0.816	0.127	-0.406
Cox-MLP (DeepSurv)	0.841	0.111	-0.349
Cox-MLP (CC)	0.844	0.119	-0.379
Cox-Time	0.861	**0.107**	**-0.334**
DeepHit	**0.888**	0.147	-0.489
RSF (Mortality)	0.855	0.112	-0.352
RSF (C^{td})	0.870	0.111	-0.352

Table 7: Evaluation metrics for the KKBox data.

a grid search over a small number of suitable parameters. The hyperparameter search is described in detail in Appendix A.2.

The best configurations are given in Table 6. For RSF, the hyperparameter search based on C^{td} yielded 8 covariates sampled for each split and a minimum of 50 observations in each leaf node. With concordance of mortality as the validation criterion, the best fitted model used 2 covariates for splitting, and a minimum leaf node size of 10. Furthermore, we found that 500 trees were sufficient, as there was little improvement compared to 250 trees.

6.2.2 Results

For evaluation, we fitted each of the methods five times and computed the time-dependent concordance index (C^{td}), the integrated Brier score (IBS), and the integrated binomial log-likelihood (IBLL) of the fitted models. The median scores are presented in Table 7. We use the median because the two proportional Cox-MLP methods yielded rather unstable results, where some of the fitted models performed very badly.

From the table, we see that DeepHit continues to outperform the other methods in terms of concordance while having the worst performance in terms of IBS and IBLL. Furthermore, Cox-Time has the best IBS and IBLL, while still providing a decent concordance. RSF continues to do well across all metrics, while again, the tuning based on C^{td} seems to yield better results than tuning base on the concordance of the mortality. Cox-SGD (Linear) does rather poorly, as it is very restricted, and serves more as a baseline in this context. The Cox-MLP methods seem to again perform reasonably close to each other, at least when taking into account that we found both methods to yield rather unstable results. We are

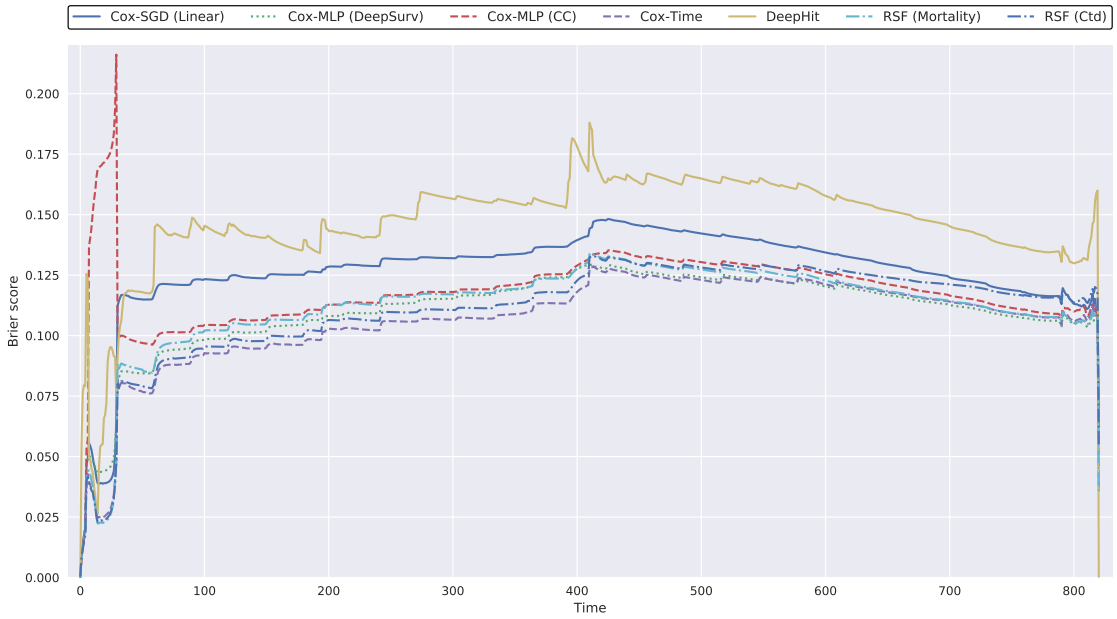


Figure 4: Brier score on KKBox data set. The methods are the same as in Table 7.

not sure why this was the case, but note that the combination of the flexible neural net and the proportionality constraint might be problematic for large data sets.

In Figure 4, we display the Brier scores used to obtain the IBS. Again, we see that DeepHit does poorly for all durations. The instability of the Cox-MLP (CC) is also very apparent for shorter durations. Cox-Time is clearly doing very well for all durations, but interestingly, we observe the Cox-MLP (DeepSurv) provides the best fitted model for larger durations. We could make a corresponding figure for the binomial log-likelihood, but as it is very similar to the Brier score plot, and provide us with no new insights, we have not included it.

6.2.3 Survival Curves

One of the benefits survival models have over binary classifiers is their ability to produce survival curves. In Figure B.1 in Appendix B, we show nine examples of estimated survival curves from Cox-Time on the test data. The curves nicely illustrate the extent of detail the method has learned.

To obtain a more general view of the predictions, we cluster the estimated survival curves of the test set. For an equidistant grid $0 = \tau_0 < \tau_1 < \cdots < \tau_m$, the survival curve of individual i is represented by a vector $[\hat{S}(\tau_0 \,|\, \mathbf{x}_i), \hat{S}(\tau_1 \,|\, \mathbf{x}_i), \ldots, \hat{S}(\tau_m \,|\, \mathbf{x}_i)]$, and by considering these as feature vectors, we apply K-means clustering to the test set with 10 clusters. In Figure 5, we display the cluster centers and the proportions of the test set assigned to each of the clusters. This is a reasonable approach for segmenting customers based on their churning behavior. A natural next step would be to further investigate the clusters, but as we consider this somewhat outside the scope of this paper, we only make

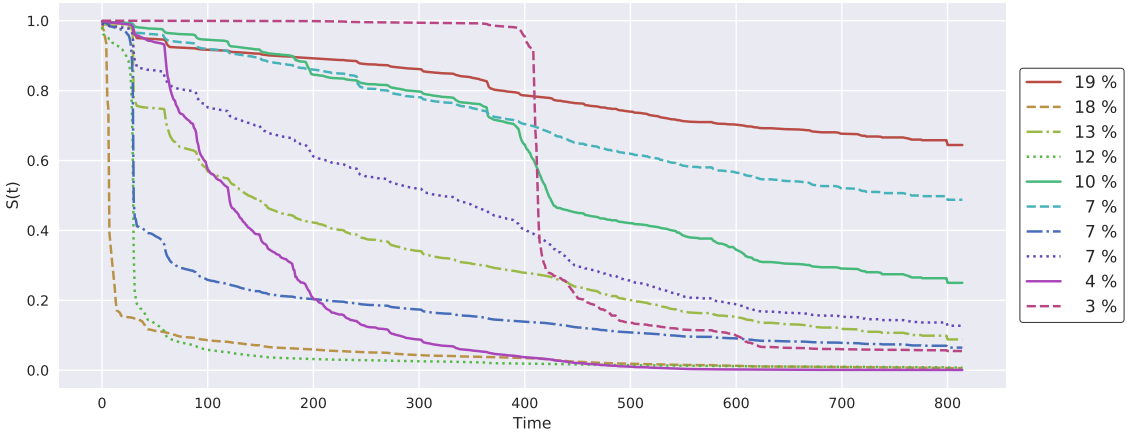


Figure 5: Cluster centers of survival curves from KKBox test data. The centers were generated by K-means clustering on survival curves from Cox-Time. The legend gives the proportion of test data assigned to each cluster (rounded to nearest integer).

a few observations. First, we see that 19 % of the customers are assigned to a cluster that does not provide much detailed information about their behavior, but instead provides a survival curve with a rather constant slope. In sharp contrast, the second largest group (18 %) is at high risk of churning immediately. Furthermore, we observe that many of the curves seem to have higher hazards at the end of each month (drops in the survival curves around 30, 60, 90 days), and we hypothesize that this is a result of customers paying for a full month at a time.

The smallest cluster, constituting only 3 % of the test data, has a sharp drop around day 400. Investigating the covariates of the assigned customers reveals that most of them had prepaid for a 411 days long subscription. However, the large drop after 400 days indicates that only around 25 % of them were interested in continuing their subscription.

Our choice of 10 clusters is mainly motivated by how many curves that can be visualized in a single plot, without being too crowded. Further increasing the number of clusters would likely reveal more detailed behavior.

7. Discussion

In this paper, we propose extensions of the Cox proportional hazards model. By parameterizing the relative risk function of a Cox model with neural networks, we can model rich relationships between the covariates and event times. Furthermore, we allow the networks to model interactions between the covariates and time, resulting in models that are no longer constrained by the proportionality assumption of the Cox model. Building on methods for nested case-control studies, we propose a loss function that can be computed in batches, enabling the models to scale to large data sets. We verify through simulation studies that fitting a Cox model using our proposed loss function gives results close to those obtained using the full Cox partial likelihood.

We compare our suggested methodology with classical Cox regression, random survival forests (RSF), DeepHit, and DeepSurv on 5 real-world data sets, and find that our proposed Cox-Time method performs very well, and has the best overall performance in terms of integrated Brier score (IBS) and integrated binomial log-likelihood (IBLL). DeepHit has, in general, the best discriminative performance, but this comes at the cost of poorly calibrated survival estimates.

Finally, we show how estimated survival curves (event probabilities as functions of time), can be used as a descriptive tool to better understand event-time data sets. This is illustrated by an example where we cluster the survival estimates of a customer churn data set, and show that this customer segmentation provides a useful view of the churning process.

Interesting expansions of our methodology include the extension to handle multiple competing events, time-dependent covariates, dynamic predictions, and recurrent events. Furthermore, it would be of interest to explore other data sources that require more advanced network structures such as convolutions and recurrent neural networks. Finally, less computationally expensive alternatives for creating survival estimates for the Cox-Time method should be explored.

Acknowledgments

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Appendix A. Details on Hyperparameter Tuning

In the following, we provide further details about the experiments conducted in Section 6. Here we list hyperparameter configurations and details about the model fitting procedures.

A.1 Four Common Survival Data Sets Tuning

Table A.1 gives the hyperparameter search space used for Rot. & GBSG, SUPPORT, METABRIC, and FLCHAIN. The square brackets describe continuous variables. In the experiments in Section 6.1, we sample 300 random parameter configurations for each method, for each fold of each data set. In the table, "α" and "σ" are given in (D.2) in Appendix D, "Num. durations" are the number of discrete durations (equidistant) used in DeepHit, "λ" is the penalty in (10), "Log duration" refers to a log-transform of the durations passed to Cox-Time, "Ridge" is a ridge penalty used in classical Cox regression, "Split covariates" and "Size leaf" are the number of covariates used for splitting, and the minimum node size of RSF.

A.2 KKBox Tuning

Hyperparameters in the KKBox study were found by a grid search over the relevant parameters in Table A.2. The table consists of three sections, where the top represents the networks, the bottom represents RSF, and the middle contains network parameters that were found on a smaller network with two layers and 128 nodes. "α" and "σ" controls the